

Ambr 15 Cell Culture Generation 2 - 10-15 mL working volume

sartorius-ambr-15

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	10 mL	Source-supported
Maximum working volume	15 mL	Source-supported
Nominal vessel / bag size	15 mL	Source-supported
Total vessel volume loaded into the form	18 mL	Source-supported
Initial working volume loaded into the form	10 mL	Derived / estimated
Product-contact material	Polycarbonate / polyethylene	Source-supported
Height : diameter	2	App assumption
Internal / chamber diameter	0.0212 m	Derived / estimated
Liquid height at maximum working volume	0.0424 m	Derived / estimated
Impeller / mixing type	Pitched-blade turbine	Source-supported
Impeller count	1	Source-supported
Impeller diameter	0.0114 m	Source-supported
Power number	2.15	Source-supported
Initial agitation	800 rpm	App assumption
Maximum agitation	2500 rpm	Source-supported
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.06666667 vvm	App assumption
Baffled vessel	No	App assumption
Maximum pressure	0.2 bar(g)	App assumption
Default operating pressure	0 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 10 mL

Source-supported | Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt; Microbioreactor vessel table

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 10 mL.

Limitations: User-supplied screenshot excerpt from Scribd; Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification. Revision, publication date and page number unknown; not a complete original PDF.

Local original: [Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt - screenshot excerpt; page unknown](#)

[Third-party document listing \(Scribd\)](#)

Maximum working volume - 15 mL

Source-supported | Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt; Microbioreactor vessel table

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 15 mL.

Limitations: User-supplied screenshot excerpt from Scribd; Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification. Revision, publication date and page number unknown; not a complete original PDF.

Local original: [Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt - screenshot excerpt; page unknown](#)

[Third-party document listing \(Scribd\)](#)

Nominal vessel / bag size - 15 mL

Source-supported | Sartorius Ambr 15 Cell Culture Generation 2 brochure - PDF page 10; Functions and micro bioreactor vessel

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 15 mL.

Limitations: Generation 2 cell-culture configuration; stirring range 150-2000 rpm.

Local original: [Sartorius Ambr 15 Cell Culture Generation 2 brochure - PDF page 10](#)

[Manufacturer publication](#)

Total vessel volume loaded into the form - 18 mL

Source-supported | Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt; Microbioreactor vessel table

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 18 mL.

Limitations: User-supplied screenshot excerpt from Scribd; Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification. Revision, publication date and page number unknown; not a complete original PDF.

Local original: [Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt - screenshot excerpt; page unknown](#)

[Third-party document listing \(Scribd\)](#)

Initial working volume loaded into the form - 10 mL

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(10 mL, 15 mL) = 10 mL

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Polycarbonate / polyethylene

Source-supported | Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt; Microbioreactor vessel table

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Polycarbonate / polyethylene.

Limitations: User-supplied screenshot excerpt from Scribd; Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification. Revision, publication date and page number unknown; not a complete original PDF.

Local original: Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt - screenshot excerpt; page unknown

Third-party document listing (Scribd)

Height : diameter - 2

App assumption | Generic microbioreactor profile

Rule: PROFILE_DEFAULTS["micro"].heightDiameterRatio = 2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbioreactor starting profile. This ratio is subsequently used to estimate vessel diameter and liquid height.

Internal / chamber diameter - 0.0212 m

Derived / estimated | Cylindrical geometry estimate

Rule: $D = \text{cubert}[(4 \times V)/(\pi \times H:D)]$

Selection / calculation: $D = \text{cubert}[(4 \times 0.000015 \text{ m}^3)/(\pi \times 2)] = 0.02121569 \text{ m}$; the form displays 0.0212 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 0.0424 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.02121569 \text{ m} \times 2 = 0.04243138 \text{ m}$; the form displays 0.0424 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Pitched-blade turbine

Source-supported | Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt; Microbioreactor vessel table

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Pitched-blade turbine.

Limitations: User-supplied screenshot excerpt from Scribd; Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification. Revision, publication date and page number unknown; not a complete original PDF.

Local original: Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt - screenshot excerpt; page unknown

Third-party document listing (Scribd)

Impeller count - 1

Source-supported | Sartorius Ambr 15 Cell Culture Generation 2 brochure - PDF page 10; Functions and micro bioreactor vessel

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.

Limitations: Generation 2 cell-culture configuration; stirring range 150-2000 rpm.

Local original: Sartorius Ambr 15 Cell Culture Generation 2 brochure - PDF page 10

Manufacturer publication

Impeller diameter - 0.0114 m

Source-supported | Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt; Microbioreactor vessel table

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.0114 m.

Limitations: User-supplied screenshot excerpt from Scribd; Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification. Revision, publication date and page number unknown; not a complete original PDF.

Local original: Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt - screenshot excerpt; page unknown

Third-party document listing (Scribd)

Power number - 2.15

Source-supported | Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt; Microbioreactor vessel table

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2.15.

Limitations: User-supplied screenshot excerpt from Scribd; Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification. Revision, publication date and page number unknown; not a complete original PDF.

Local original: Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt - screenshot excerpt; page unknown

Third-party document listing (Scribd)

Initial agitation - 800 rpm

App assumption | Generic microbioreactor profile

Rule: PROFILE_DEFAULTS["micro"].baseRpm = 800

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbioreactor starting profile. This is the initial agitation setting loaded into the form.

Maximum agitation - 2500 rpm

Source-supported | Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - operating screenshot excerpt; System operating parameters: agitation speed 150-2500 rpm

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2500 rpm.

Limitations: User-supplied screenshot excerpt from Scribd; Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification. Revision, publication date and page number unknown; not a complete original PDF. Conflicts with archived brochure PDF p. 10 (150-2000 rpm). Technical-specification excerpt selected by user; edition chronology and installed-hardware applicability unresolved.

Local original: Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - operating screenshot excerpt - screenshot excerpt; page unknown

Third-party document listing (Scribd)

Initial aeration - 0.05 vvm

App assumption | Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - operating screenshot excerpt; Maximum air or total gas flow

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0.05 vvm.

Limitations: Selected starting aeration below the conservative constant ceiling; not a manufacturer setpoint.

Local original: Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - operating screenshot excerpt - screenshot excerpt; page unknown

Third-party document listing (Scribd)

Maximum aeration - 0.06666667 vvm

App assumption | Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - operating screenshot excerpt; Maximum air or total gas flow

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 0.06666667 vvm.

Limitations: Published maximum total gas flow is 1 mL/min. Constant simulator ceiling selected at 15 mL: 1/15 vvm. At 10 mL the physical flow limit corresponds to 0.1 vvm; the app does not enforce a volume-dependent absolute-flow cap. User-supplied screenshot excerpt from Scribd; Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification. Revision, publication date and page number unknown; not a complete original PDF.

Local original: [Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - operating screenshot excerpt - screenshot excerpt](#); page unknown

[Third-party document listing \(Scribd\)](#)

Baffled vessel - No

App assumption | Generic microbioreactor profile

Rule: `PROFILE_DEFAULTS["micro"].baffled = false`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbioreactor starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.2 bar(g)

App assumption | Generic microbioreactor profile

Rule: `PROFILE_DEFAULTS["micro"].maxPressure = 0.2`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbioreactor starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0 bar(g)

App assumption | Generic microbioreactor profile

Rule: `PROFILE_DEFAULTS["micro"].operatingPressure = 0`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbioreactor starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: `maxOxygenFraction = 100`

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

This Ambr 15 preset is a cell-culture microbioreactor archetype rather than a high-oxygen-demand microbial fermenter.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

User-supplied screenshot excerpt from Scribd; Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification. Revision, publication date and page number unknown; not a complete original PDF.

Internal L x W x H = 28.0 x 14.6 x 59.7 mm. These are non-cylindrical vessel dimensions, not a tank diameter or working liquid height. Existing cylinder geometry remains an unverified simulation approximation.

kLa benchmark: 17.6 per hour, sparged vessel, 13 mL DI water, 1500 rpm, 1 mL/min gas. Reference condition only; no universal kLa calibration or RPM rating inferred.

Operating excerpt: agitation 150-2500 rpm; maximum air or total gas 1 mL/min. Brochure says 150-2000 rpm; source conflict unresolved. Existing app has no minimum-RPM or fill-dependent gas-flow enforcement.

Standard culture temperature 33-40 C +/-0.5 C (+8 C above ambient); cooled 20-40 C +/-0.5 C; temperature shift ≥ 5 C per 30 minutes. pH setpoint 6.5-7.5, monitoring 6.0-8.0 +/-0.1; DO monitoring 0-200% air saturation, accuracy +/-2% at 100%. Monitoring ranges are not control or inlet-gas limits.

No pressure rating is stated in these excerpts; retained pressure ceiling remains an unverified app assumption. Cover excerpt: sartorius-ambr15-gen2-excerpt-cover-a4e58124000c.png

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - vessel screenshot excerpt

sartorius-ambr15-gen2-excerpt-vessel-c41555e1d88b.png

SHA-256: c41555e1d88ba51ea26b764cfdd1481cd33ee1281b5bfcd4b1c2919c707068f5

Sartorius Ambr 15 Cell Culture Generation 2 brochure

sartorius-ambr-15-cell-culture-generation-2-brochure-12c854335cda.pdf

SHA-256: 12c854335cdac4c797b0ff93eb03774ee95510c4380695d5287e3cf4e8cc1672

Sartorius Ambr 15 Cell Culture Generation 2 Technical Specification - operating screenshot excerpt

sartorius-ambr15-gen2-excerpt-operating-5f9b2aced420.png

SHA-256: 5f9b2aced4209ac22914dffa4c3ef0b94b185e21b661cd0787789af048a2ecfb